

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404837
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Langvardt Krogh; Lars et al.

Powertrain assembly for a wind turbine

Abstract

A wind turbine powertrain assembly including a gearbox coupled to an electrical power generator for a wind turbine. The power generator comprises a stator at a radially outward position and a rotor in a radially inward position, wherein the rotor comprises a cylindrical field structure that is configured to rotate about a generator rotor axis, and wherein the rotor is configured to define a central hollow portion. The rotor further comprises a rotor support frame connected to and structurally supporting the cylindrical field structure, wherein the rotor support frame comprises: a rotor connection flange attaching the rotor support frame to the cylindrical field structure, a gearbox connecting flange connecting the rotor support frame to an output drive shaft of the gearbox, and a transition section extending between the rotor connection flange and the gearbox connection flange, the transition section defined at least in part by one or more stepping regions, the or each stepping region shaped to define a surface that is inclined at an angle of less than 30 degrees to the rotor axis. Advantageously, the stepping regions provide an integrated platform inside the hollow portion on which service personnel can stand to work inside the generator.

Inventors:	Langvardt Krogh; Lars (Egå, DK), Mannik; Henrik Zaar (Randers SV, DK), Broge; Rolf Nyborg (Tilst, DK), Gündüz; Mustafa (Lübeck, DE)
Applicant:	Vestas Wind Systems A/S (Aarhus N., DK)
Family ID:	80930329
Assignee:	Vestas Wind Systems A/S (Aarhus N., DK)
Appl. No.:	18/578045
Filed (or PCT Filed):	March 10, 2022
PCT No.:	PCT/DK2022/050042
PCT Pub. No.:	WO2023/001347
PCT Pub. Date:	January 26, 2023

Prior Publication Data

Document Identifier

US 20240318636 A1

Publication Date

Sep. 26, 2024

Foreign Application Priority Data

DK

PA 2021 70387

Jul. 20, 2021

Publication Classification

Int. Cl.: **F03D15/00** (20160101); **F03D9/25** (20160101); **F03D13/10** (20160101); **H02K7/116** (20060101); **H02K7/18** (20060101)

U.S. Cl.:

CPC **F03D15/00** (20160501); **F03D9/25** (20160501); **F03D13/108** (20230801); **H02K7/116** (20130101); **H02K7/1838** (20130101); **F05B2260/301** (20130101); **F05B2260/40311** (20130101)

Field of Classification Search

CPC: F03D (15/00); F03D (9/25); F03D (13/108); H02K (7/116); H02K (7/1838); F05B (2260/301); F05B (2260/40311)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6285090	12/2000	Brutsaert et al.	N/A	N/A
2013/0264167	12/2012	Geiken	192/135	F03D 80/00
2016/0218575	12/2015	Pabst et al.	N/A	N/A
2022/0018335	12/2021	Krogh	N/A	F03D 9/25

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2617994	12/2012	EP	N/A
2698529	12/2013	EP	N/A
2020125884	12/2019	WO	N/A
2020221403	12/2019	WO	N/A
2021 08466	12/2020	WO	N/A

OTHER PUBLICATIONS

International Searching Authority, International Search Report and Written Opinion issued in corresponding PCT Application No. PCT/DK2022/050042 dated Jun. 22, 2022. cited by applicant
Danish Patent and Trademark Office, technical examination issued in DK Application No. PA 2021 70387. Dated Jan. 25, 2022. cited by applicant

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a powertrain assembly, particularly for a wind turbine.

BACKGROUND TO THE INVENTION

(2) Wind turbines convert kinetic energy from the wind into electrical energy, using a large rotor with a number of rotor blades. A typical Horizontal Axis Wind Turbine (HAWT) comprises a tower, a nacelle on top of the tower, a rotor hub mounted to the nacelle and a plurality of wind turbine rotor blades coupled to the rotor hub. Depending on the direction of the wind, the nacelle and rotor blades are turned and directed into an optimal direction by a yaw system for rotating the nacelle and a pitch system for rotating the blades.

(3) The nacelle houses many functional components of the wind turbine, including for example a main rotor shaft, gearbox and generator, as well as convertor equipment for converting the mechanical energy at the rotor into electrical energy for provision to the grid. The gearbox steps up the rotational speed of the low-speed main shaft and drives a gearbox output shaft. The gearbox output shaft in turn drives the generator, which converts the rotation of the gearbox output shaft into electricity. The electricity generated by the generator may then be converted as required before being supplied to an appropriate consumer, for example an electrical grid distribution system. So-called “direct drive” wind turbines that do not use gearboxes are also known. In a direct drive wind turbine, the generator is directly driven by the main rotor shaft.

(4) Ordinarily, the generator of a wind turbine is an IPM (interior permanent magnet) electric machine composed of an external stator assembly which surrounds an internal rotor assembly. The IPM internal rotor assembly is typically composed of a ring like structure comprising multiple annular permanent magnetic packages, supported on a central shaft. The gearbox output shaft interfaces with the central shaft of the rotor assembly.

(5) WO2020143888 A1 shows a particular type of IPM electric machine for use as a generator in a wind turbine. In this example, the ring-shaped structure is supported on one of its ends by an annular support frame. Not having a central hub in the rotor assembly results in a number of important benefits, such as reduced cost and weight and improved cooling airflow. Cooling air that is provided centrally to the generator can freely flow in axial and radial directions and effectively cool the rotor and any generator parts located in its direct vicinity. Further refinements to this design are desirable, however, for example to enhance maintainability of the assembly.

(6) It is an object of the present invention to provide a solution to one or more of the problems mentioned above.

SUMMARY OF THE INVENTION

(7) According to a first aspect of the invention, there is provided a wind turbine powertrain assembly including a gearbox coupled to an electrical power generator for a wind turbine.

(8) The power generator comprises a stator at a radially outward position and a rotor in a radially inward position, wherein the rotor comprises a cylindrical field structure that is configured to rotate about a generator rotor axis, and wherein the rotor is configured to define a central hollow portion. The rotor further comprises a rotor support frame connected to and structurally supporting the cylindrical field structure, wherein the rotor support frame comprises: a rotor connection flange attaching the rotor support frame to the cylindrical field structure, a gearbox connecting flange connecting the rotor support frame to an output drive shaft of the gearbox, and a transition section

extending between the rotor connection flange and the gearbox connection flange, the transition section defined at least in part by one or more stepping regions, the or each stepping region shaped to define a surface that is inclined at an angle of less than 30 degrees to the rotor axis.

(9) An advantage of the invention is that the powertrain assembly includes a gearbox coupled to a generator which is configured particularly to ease the process of servicing. Typically, powertrain systems including gearboxes that are coupled to generators require that all servicing be carried out from access points outside of the generator. However, in the powertrain assembly of the invention, components of the rotor define a central hollow portion that is suitable for being accessed by maintenance personnel for various service actions associated with the generator. What is more, the stepping regions provide an integrated platform inside the hollow portion on which service personnel can stand to work inside the generator.

(10) The or each stepping region may extend circumferentially around the rotor axis, either fully or partially. The stepping region can be a single continuous surface or can be defined by a plurality of separate stepping regions. In the case of a plurality of stepping regions, at least some of them may be separated from each other by a strengthening rib that provides structural strength to the rotor support structure.

(11) Beneficially, the transition section that defines the stepping regions may be provided with an axially-facing end surface which may be generally aligned along the direction of the rotor axis with the gearbox connection flange. The end surface may define a plurality of access apertures that permit a service engineer to access areas outside of the rotor support frame during servicing actions.

(12) In some embodiments, the rotor may be housed in a generator service cabinet, which may be connected to a gearbox housing. The generator service cabinet may define an internal chamber in which the rotor is received, and may be provided with one or more closure elements which are configured to selectively close an opening defined by the generator service cabinet into the internal chamber. The service cabinet therefore results in a normally closed environment for servicing purposes which service engineers may access when required. Suitable access control system may be associated with the closure to ensure that access is only enabled under safe conditions. The closed environment therefore means that the service area can be more readily kept clear of contamination.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention will now be described, by way of example only, with reference to the attached drawings, in which:
- (2) FIG. 1 is a front view schematic diagram showing a known wind turbine configuration;
- (3) FIG. 2 is a view of example powertrain components that may be housed in a nacelle of the wind turbine in FIG. 1;
- (4) FIG. 3 is a cutaway view of a known configuration including a generator coupled to a gearbox, shown in axially-spaced positions;
- (5) FIG. 4 is a perspective view of a generator rotor assembly of the known generator in FIG. 3, as seen from a non-drive end thereof;
- (6) FIG. 5 is a perspective view of the rotor assembly shown in FIG. 4, but seen from a drive-end;
- (7) FIG. 6 is a longitudinal cross-section of a generator rotor assembly in accordance with an embodiment of the invention, viewed in part-schematic form located inside a generator service cabinet. The generator rotor assembly comprises a cylindrical field structure that is mated to a rotor support hub or frame and may be used in the generator as shown in the previous Figures;
- (8) FIG. 7 is a perspective view of a generator rotor assembly of FIG. 6 in isolation of the generator

service cabinet;

(9) FIG. **8** is another front perspective view of the generator rotor assembly of FIG. **6**, but with the cylindrical field structure omitted;

(10) FIG. **9** is perspective view of the generator rotor assembly of FIG. **7**, when viewed from its non-drive end;

(11) FIG. **10** is an end view of the generator rotor assembly of FIGS. **6** to **9** inside its respective generator cabinet, showing a maintenance engineer standing inside the support frame in order to access internal components; and

(12) FIGS. **11a** and **11b** show end views like that in FIG. **10** but with an access door in closed and open conditions, respectively.

DETAILED DESCRIPTION

(13) A specific embodiment of the present invention will now be described in which numerous features will be discussed in detail in order to provide a thorough understanding of the inventive concept as defined in the claims. However, it will be apparent to the skilled person that the invention may be put into effect without the specific details and that in some instances, well known methods, techniques and structures have not been described in detail in order not to obscure the invention unnecessarily.

(14) The invention relates in general to an electrical generator configuration which has several advantages associated with maintainability of the generator and the associated gearbox to which it is connected. In particular, the generator configuration provides a maintenance engineer with improved access inside the generator which enables various types of adjustments to be made without disassembly of some or all of the generator, which is a significant advantage in the confines of a wind turbine nacelle.

(15) In order to place the embodiments of the invention in a suitable context, reference will firstly be made to FIG. **1**, which illustrates a typical Horizontal Axis Wind Turbine (HAWT) in which a generator rotor assembly according to an embodiment of the invention may be implemented. Although this particular image depicts an on-shore wind turbine, it will be understood that equivalent features will also be found on off-shore wind turbines. In addition, although the wind turbines are referred to as 'horizontal axis', it will be appreciated by the skilled person that for practical purposes, the axis is usually slightly inclined to prevent contact between the rotor blades and the wind turbine tower in the event of strong winds.

(16) The wind turbine **1** comprises a tower **2**, a nacelle **4** rotatably coupled to the top of the tower **2** by a yaw system, a rotor hub **8** mounted to the nacelle **4** and a plurality of wind turbine rotor blades **10** coupled to the rotor hub **8**. The nacelle **4** and rotor blades **10** are turned and directed into the wind direction by the yaw system.

(17) The nacelle **4** houses many functional components of the wind turbine, including the generator, gearbox, and rotor brake assembly, as well as convertor equipment for converting the mechanical energy of the wind into electrical energy for provision to the grid. FIG. **2** illustrates an example of a layout within the nacelle **4**, that layout including a main shaft **26**, which extends through a main bearing housing **20**, a gearbox **22** and a generator **24**. The main shaft **26** is connected to, and driven by, the rotor **8** and provides input drive to the gearbox **22**. Collectively, these components may be considered to be a powertrain of the wind turbine. The gearbox **22** steps up the rotational speed of the low-speed main shaft via internal gears (not shown) and drives a gearbox output shaft. The gearbox output shaft in turn drives the generator **24**, which converts the rotation of the gearbox output shaft into electricity. The electricity generated by the generator **24** may then be converted by other components (not shown) as required before being supplied to an appropriate consumer, for example an electrical grid distribution system.

(18) The gearbox **22** and generator **24** may be coupled together in an integrated unit. FIG. **3** shows the generator **24** in more detail. With reference firstly to the gearbox **22**, a gearbox housing is generally cylindrical in form which is due to the specific type of gearbox that is used in the

illustrated embodiment, which is an epicyclic gearbox. As the skilled person would know, an epicyclic gearbox comprises a series of planet gears that are arranged about a central sun gear, and which collectively are arranged within an encircling ring gear. The general configuration of the gearbox is such that the input shaft and output shaft are arranged on a common axis. The ratio of the number of teeth between the ring gear, the planet gear and the sun gears determines the gear ratio of the gearbox. For clarity, fine detail of the gearbox will not be described in further detail here as the gearbox is not the principal subject of the invention. Suffice to say that other gearbox configuration could also be used, although it is currently envisaged that an epicyclic gearbox provides an elegant solution fit for the confines of a wind turbine nacelle. Such an epicyclic gearbox may contain more than one epicyclic stage, for example one, two or three epicyclic gear stages.

(19) Referring still to the cutaway view of FIG. 3, but also to FIGS. 4 and 5, the generator 24 is an IPM (interior permanent magnet) electric machine having an external stator assembly 30, which surrounds an internal rotor assembly 32. The stator assembly 30 includes stator windings 38, a stator core 40 and a stator frame (not shown) which surrounds and supports the stator windings 38 and stator core 40. It is however noted that the invention is not limited to a specific type of stator. Notably, the internal configuration of the generator 24 shown in FIG. 3 is provided here by way of example and to give the invention its proper context. The specific configuration of the generator rotor 32 in FIGS. 3 to 5 does not form part of the invention and is provided for a better understanding of the configuration of a generator rotor in accordance with an embodiment of the invention, as will be described later with reference to FIGS. 6 to 11.

(20) The gearbox 22 is coupled to the generator 24 by way of an output shaft 31 which connects to the rotor assembly 32 of the generator 24. As such, the major axis of the gearbox output shaft 31 defines the rotational axis of the generator 24. Notably, the gearbox also includes an input shaft 33 which is coaxially aligned with the gearbox output shaft 31. Although the generator rotor assembly 32 is shown in cross section in FIG. 3 in-situ within the generator 22 it is shown in isolation in FIGS. 4 and 5.

(21) The generator rotor assembly 32 has a drive end 34 and a non-drive end 36. The drive end 34 faces towards the gearbox 22 and the non-drive end 36 faces away from the gearbox 22. The non-drive end 36 of the generator rotor assembly 32 can be seen in FIG. 4, and the drive-end 34 of the generator rotor assembly 32 can be seen in FIG. 5.

(22) The generator rotor assembly 32 comprises a cylindrical ring structure 40 that defines a hollow central region and which is arranged to rotate around a rotational axis. The cylindrical ring structure 40 comprises a plurality of solid circular rotor bars 42 which accommodate permanent magnet packages. The ring structure 40 therefore is responsible for generating the rotating magnetic field of the generator, in use. Here, the bars 42 are shown as being of equal circumference and thickness, but that can vary. The bars 42 are arranged coaxially around the rotational axis, such that when assembled the arrangement of bars 42 defines a cylindrical structure with a central hollow area. The bars 42 are spaced apart by an equal distance such that a gap is defined in between each pair of bars 42.

(23) These gaps allow air that is provided centrally to the generator to flow through the rotor structure and cool the generator rotor assembly as well as other parts of the generator, including parts that are located radially outside the rotor assembly 32. This airflow is further enhanced by the fact that no central hub is needed for providing structure and support for the rotor assembly 32.

(24) The cylindrical ring structure 40 is connected to a rotor support frame 50. The rotor support frame 50 comprises a radially outer first connection flange 52, which is best seen in FIG. 4, and a radially inner second connection flange 54, which is best seen in FIG. 5. The first connection flange 52 serves to connect the bars 42 to the rotor support frame 50, and so can be considered to be a 'rotor connection flange', whilst the second connection flange 54 serves to couple the rotor support frame 50 to the gearbox output shaft 31, and so can be considered to be a 'gearbox

connection flange'. In FIG. 5, a set of connecting bolts **56** can be seen extending perpendicularly out of the second connecting flange **54** which would be used to fix the support frame **50** to the gearbox output shaft **31**.

(25) The support frame **50** and the cylindrical ring structure **40** are connected together by a plurality of tie rods **58** (only two of which are identified in FIGS. 4 and 5) which extend axially through respective holes in the bars **42** and are secured through the rotor connection flange **52** of the support frame **50**. Suitable mechanical fasteners such as bolts are provided on extremities of each of the tie rods **58** and are tightened to place the package of bars **42** into compression thus securing the cylindrical ring structure **40** securely to the support frame **50** as an integral unit.

(26) It will be noted that the rotor connection flange **52** and the gearbox connection flange **54** are axially-facing surfaces and extend in respective planes that are parallel to but spaced apart from one another along the rotor axis. In this respect, the gearbox connection flange **54** extends somewhat into the hollow internal area defined by the cylindrical ring structure. Moreover, the circumference of the gearbox connection flange **54** is smaller than the rotor connection flange **52**.

(27) The support frame **50** further comprises a transition section **60** that extends between the rotor connection flange **52** and the gearbox connection flange **54**. The transition section **60** is generally frustoconical and extends at a steep angle with respect to the rotor axis. As shown here, the transition section **60** defines an angle of approximately 70 to 80 degrees with respect to the rotor axis. In other words, the transition section **60** has a cone angle of approximately 140 to 160 degrees.

(28) When the generator is assembled, the generator rotor assembly **32** is surrounded by the external stator **30**. In turn, both the rotor assembly **32** and the stator **30** are surrounded by a generator housing or cabinet **68**, which can be best appreciated by FIG. 3.

(29) With reference to the known configuration of generator rotor assembly **32** as described above, it should be appreciated that the configuration provides various advantages in terms of generator design and efficiency, but it also presents challenges. For example, interior access of the generator is limited which makes it problematic for maintenance engineers to access the internals of the generator in order to check internal components and take remedial action where required.

(30) The embodiments of the invention involved an improved configuration of generator rotor assembly that address these issues. An embodiment of the invention will now be described with reference to the remaining figures.

(31) The illustrated embodiment provides a generator rotor assembly **70** which is equivalent in function to the known configuration of generator rotor assembly **32** discussed above. However, the embodiments of the invention confer several advantageous aspects, as will be explained in further detail in this discussion.

(32) Whereas FIG. 6 illustrates the generator rotor assembly **70** in part-schematic view in relation to the surrounding components of the generator housing or 'cabinet' **81**, FIGS. 7 to 9 show the generator rotor assembly **70** in isolation from the generator cabinet **81**. At this point, it should be noted that the generator rotor assembly **70** of the illustrated embodiment shares many similarities to that discussed earlier so only the differences will be focused on here.

(33) At a broad level, the generator rotor assembly **70** includes a rotor support frame **72** which supports a cylindrical field structure **74** that acts as a magnet carrier thereby to generate a rotating magnetic field in use. The configuration of the cylindrical field structure **74** is comparable to what has been described with reference to FIGS. 3 to 5, so further detailed discussion on this assembly will be omitted.

(34) The generator rotor assembly **70** includes a drive end **76**, and a non-drive end **78**. The drive end **76** is connectable to an output drive shaft **77** of the gearbox **22** and the non-drive end **78** is distal from the gearbox **22**. In effect, the generator rotor assembly **70** can be considered to be rotationally suspended from the gearbox output shaft **77** in a cantilever manner as the non-drive end is not rotationally supported by a bearing.

(35) In essence, the rotor support frame **72** provides a means of coupling the gearbox output shaft **77** to the cylindrical field structure **74**. To this end, the rotor support frame **72** comprises a gearbox connection flange **80** and a rotor connection flange **82**.

(36) The gearbox connection flange **80** and the rotor connection flange **82** are both axial-facing surfaces that are oriented perpendicular to the rotational axis R of the generator rotor assembly **70** and are spaced along that axis relative to one another. Both flanges **80,82** therefore extend in a plane perpendicular to the rotational axis R in the illustrated embodiment.

(37) The gearbox connection flange **80** comprises an annular fixing slot arrangement **84** through which means the rotor support frame **70** is connectable to the gearbox output shaft **77** by a set of suitable bolts (not shown). The fixing slot arrangement **84** therefore defines a first pitch circle diameter which is shown in FIG. **10** as “D1”.

(38) The rotor connection flange **82** is coaxial with the gearbox connection flange **80** and has a larger diameter. The rotor connection flange **82** secures to the cylindrical field structure **74** by means of a tie-rod system **86** comprising a plurality of circumferentially spaced tie rods **88** that extend through the individual bars of the cylindrical field structure **74**, as in the previous embodiment. The circular arrangement of tie rods **88** therefore define a second pitch circle diameter, shown as D2 in FIG. **6**. As can be seen the first pitch circle diameter is smaller than the second pitch circle diameter D2. In the illustrated embodiment, the first pitch circle diameter D1 is approximately 30% of the second pitch circle diameter D2.

(39) Conveniently, the tie rod system **86** includes a balancing means to adjust the rotational balance of the cylindrical field structure **74**. The balancing means can be achieved by adding shims or washers that can be captured between the rotor connection flange **82** and a tie rod fastener **89**, such as a hexagonal nut. Tensioning of the tie rods **88** can be achieved by appropriate turning of the respective fastener **89** which applies a compressive force to the cylindrical structure **74**, whereas adjustment of the rotational balance can be achieved by adding or removing balancing shims around the array of tie rods **88** until an optimum rotating balance is obtained.

(40) A transition section **90** extends between the gearbox connection flange **80** and the rotor connection flange **82**. The functionality of the transition section **90** is to provide rigidity to the rotor support frame **72**, whilst maintaining the rotor connection flange **82** and the gearbox connection flange **80** in a spaced relationship and promoting good airflow through the structure.

(41) As can be readily observed from the Figures, the transition section **90** comprises two main portions, namely a first frustoconical portion **92** and a second axial-facing surface portion **94**. The frustoconical portion **92** is proximal to the rotor connection flange **82** and the axial-facing surface portion **94** is proximal to the gearbox connection flange **80**.

(42) The frustoconical portion **92** comprises a plurality of regions **96** (not all of which are labelled on the Figures, for clarity) that are relatively flat since they are oriented at a shallow angle to the rotational axis. Since each of the regions **96** extends in an axial direction at a relatively shallow angle to the rotational axis, they provide an area which is suitable for supporting a maintenance engineer inside the rotor support frame **72**. Each of the regions **96** can therefore be considered a stepping region. In the illustrated embodiment, there are a total of nine stepping regions **96**.

(43) Each of the stepping regions **96** is demarked from its neighbouring region by a strengthening rib or web **98**. The strengthening ribs **98** contribute to the rigidity of the transition section. Here the stepping regions **96** are embodied as relatively thin plate-like sections.

(44) At this point, it should be appreciated that although the illustrated embodiment includes a plurality of stepping regions **96**, this is not essential and instead the transition section **90** may define any number of stepping regions, which includes only a single continuous stepping region that defines a curved surface extending circumferentially about the transition section **90** substantially uninterrupted.

(45) As can be seen in FIG. **11**, the stepping regions **96** provide a beneficially stable surface or platform that is flat enough to support a maintenance engineer inside an internal chamber **97** that is

surrounded by the rotor support frame **72** and, thus, also by the generator cabinet **81**. This aspect is also benefitted by various other aspects of the rotor support frame, particularly its internal dimensions. For example, the internal diameter of the rotor connection flange is greater than 1.8 m, but can be greater than 2.0 m, or even 2.2 m. This provides suitable headroom for a maintenance engineer inside the rotor support frame **72**. This aspect can be further appreciated by observing the transition section **60** in FIGS. **3** to **5**, which defines a much steeper angle with respect to the rotational axis which limits the size of the space inside the rotor assembly. Moreover, the steep surface presented by the transition section **60** would not be possible for a maintenance engineer to stand on comfortably or safely.

(46) A further beneficial aspect of the rotor support frame **72** is the axial dimension between the rotor connection flange **82** and the gearbox connection flange **80**. In the illustrated embodiment, the dimension marked as 'A' (see FIG. **6**) indicates the length along the rotational axis between the rotor connection flange **82** and the gearbox connection flange **80**. The dimension A is preferably greater than 0.5 m, to give useful 'depth' in the internal chamber **97** of the rotor support frame **72** to accommodate maintenance personnel fully inside. Still preferably, the dimension A is greater than 0.6 m.

(47) The depth in the internal chamber **97** of the rotor support frame **72** is also benefitted by the shallow angle of the stepping regions **96**. As can be seen in FIG. **6**, the stepping regions **96** define an angle of less than 30 degrees with respect to a line parallel to the rotational axis. This angle is indicated as 'B' in FIG. **6**. Expressed another way, the portion of the transition section **90** defines a cone angle of less than 60 degrees. The shallow angle that the frustoconical section defines with the rotor axis R contributes to the depth of the internal chamber **97** of the cylindrical field structure **74**.

(48) As has been discussed above, due to the internal dimensions of the internal volume **97** of the rotor support frame **70**, both its internal diameter and the depth 'A', a maintenance engineer is able to climb inside it and access various of its components and other components associated with the generator. This is a significant advantage because it means that maintainability of the generator is enhanced considerably.

(49) As well as defining the stepping regions **96**, the transition section **90** further includes the axial-facing surface portion **94**. The axial-facing surface portion **94** defines the drive end **76** of the transition section **90** as the outer surface of the transition section **90** turns through approximately 90 degrees to provide a flat end face which is substantially perpendicular to the rotational axis R. Beneficially, the strengthening ribs **98** extend along the stepping regions **96** in a direction aligned with the rotational axis and span between the stepping regions **96** and the axial-facing surface portion **94**.

(50) The axial-facing surface portion **94** is provided with an array of access apertures **100** which penetrate the depth of the axial-facing surface portion **94**.

(51) The access apertures **100** are arranged in a circular array in this embodiment and are angularly equi-spaced. In total there are nine access apertures **100** in the illustrated embodiment, arranged in a one-per-stepping region **96** configuration. The circular array of access apertures **100** is radially inwards of the stepping regions **96** and is radially outwards of the gearbox connection flange **80**. In this sense, it can be considered located radially between the gearbox connection flange **80** and the stepping regions **96**.

(52) The access apertures **100** permit access through the rotor support frame **72** so that a maintenance engineer can carry out a range of tasks such as monitoring components of the gearbox output shaft such as bearing sensors (temperature sensors and accelerometers) rotary encoders for example. To this end, the access apertures should be of a suitable cross-sectional area to allow a maintenance engineer to extend a hand through one of the apertures and reach as far as is necessary. For this purpose, it is envisaged that the access apertures each should provide an open area of at least 100 cm.^{sup.2}. Optionally a larger size would have benefits, such as at least 200 cm.^{sup.2}, as this would provide a larger reach through area whilst allowing improved visibility

through the aperture.

(53) Although the rotor support frame **70** functions as a single component to connect between and transmit drive from the gearbox output shaft to the cylindrical field structure **74**, conveniently in the illustrated embodiment the rotor support frame **72** comprises at least first and second components, labelled here respectively as **102** and **104**. Optionally, the components may be cast, e.g. of steel of a suitable grade. The first component **102** is radially inwards of the second component, and both components function together as a hub for the cylindrical field structure **74**. The first component **102** can therefore be considered an “inner hub component” **102** and the second component **104** can therefore be considered an “outer hub component” **104**.

(54) As can be seen from the Figures the gearbox connection flange **80** is part of the inner hub component **102**, being separate to the outer hub component **104** which defines the transition section **90** and the rotor connection flange **82**. The two components **102,104** are connected at a circular array **106** of bolts that is radially outwards of the access apertures **100**. Notably, the circular array **106** has a pitch circle diameter marked as D3 on FIG. 6. The pitch circle diameter D3 of the circular array **106** is greater than the pitch circle diameter D1 of the fixing array **84**. Manufacturing the rotor support frame **72** out of at least two components in this way provides a manufacturing advantage. Furthermore, this construction enables convenient access to components associated with the generator. For instance, with the rotor assembly **70** locked against rotational and radial movement, it is possible to disassemble the inner hub component **102** from the rotor support frame **72**. The inner hub component **102** can therefore be removed from the gearbox by the removing the circular array **84** of bolts connecting the rotor support frame **72** to the gearbox output shaft **77** which permits access, maintenance and removal/replacement of a bearing cassette (not shown) associated with the gearbox **22** without requiring complete disassembly of the generator.

(55) The rotor support frame **72** further comprises an extension portion **110**. The extension portion **110** extends from the outer edge of the non-drive side of the rotor connection flange **82** and is cylindrical in form. The axial dimension of the extension portion **110** is approximately half the axial length of the cylindrical field structure **74**. In the illustrated embodiment, the extension portion **110** takes the form of a thin-walled cylinder having a diameter approximately equal to the outside diameter of the rotor connection flange **82**.

(56) The non-drive end of the extension portion **110** also supports a ring gear **112** which is engaged by one or more drive motors **114**. Here, the ring gear **112** is shown bolted to the extension portion **110** at a bolt ring **111**. The ring gear **112** and drive motors **114** provide a means by which the rotational orientation of the generator rotor assembly **70** can be controlled during maintenance. For example, a maintenance engineer may need to have access to a certain part of the generator which will require the generator to be rotated to a certain position. The ring gear **112** and drive motors **114** also provide a means to cause rotation of the gearbox and main shaft via the generator rotor assembly **70**. The axial dimension of the extension portion **110** may therefore be considered to be at least partly determined based on the required position of the ring gear **112**. In the illustrated embodiment, the extension portion **110** also defines a plurality of airflow apertures **115**. The airflow apertures **115** may in principle take any form, but as shown here there are a plurality of apertures **115** that are distributed circumferentially about the extension portion **110** and which permit air to flow radially in an outward direction. The airflow apertures **115** may be configured so that they are completely open, or they may be provided with a perforated covering such as a grille to ensure that debris or loose parts are unable to pass through the apertures **115**. In addition to the airflow apertures **115** defined in the extension portion **110**, the rotor support frame **72** is also provided with a second set of airflow apertures **116** (best seen in FIGS. 8 and 9) which are located in a radially inward position compared to the first set of airflow apertures **115**. The second set of airflow apertures **116** are distributed circumferentially around the transition section **90** and in this embodiment are equi-angularly spaced. More specifically, each of the apertures **116** are located at the elbow defined where the transition section **90** bends sharply into the rotor connection flange **82**.

Each aperture **116** also features an associated guide channel **117** that extends a short way from a respective aperture **116** axially along the transition section **90**. The guide channels **117** are formed as recesses or notches in the surface of the transition section **90**. The number of second airflow apertures **116** is not crucial, but in the illustrated embodiment there are two apertures **116** for each stepping region **96**. It should be noted that the second set of airflow apertures **116** are at a radially outer position compared to the access apertures **100**. Both sets of apertures **100,116** provide airflow benefits. However, the provision of both sets of apertures **100,116** at spaced radial locations promotes a more even flow of air past the rotor support frame.

(57) The extension portion **110** also supports a brake disc **118** which is fixed to the bolt ring **111** and therefore is in a radially outer position with respect to the ring gear **112**. The brake disc **118** is acted on by a set of brake actuators **119** which are mounted on the generator service cabinet **81** and are best viewed in FIG. **10**.

(58) From the above discussion, it will be appreciated that the configuration of various aspects of the generator rotor assembly in the illustrated embodiments enhance the maintainability of various components of the generator, and particularly the generator rotor assembly. For example, the dimensions of the interior volume of the rotor support frame **72** means that a maintenance engineer can get access inside the generator. This can be useful for at least one of the following purposes. Firstly, 360 degree access is permitted to the tie rod system **86**, the associated tensioning nuts **89** and also the balancing adjustment masses **87**, such that an engineer can apply suitable adjustments to these components about the entire circumference of the rotor support frame **72** whilst in a single position inside the generator. This confers a significant advantage because adjustment can be carried out quickly without any disassembly of the machine. Furthermore, an engineer inside the generator also has 360 degree access to the circular array **106** of bolts connecting the gearbox connection flange **80** to the transition section **90**, and also 360 degree access to the circular array **84** of bolts connecting the rotor support frame **72** to the gearbox output shaft—this allows inspection of these bolted connections to be carried out quickly which is a significant advantage during maintenance inspection processes. Also, as mentioned above, this enables the inner hub component **102** to be removed so that access to the gearbox shaft **77** is achieved, thereby permitting removal and replacement of a bearing cassette. Furthermore, the significant space inside the generator allows an engineer 360 degree access to the bolted connection between the ring gear **112** and the extension portion **110**; and similar accessibility to the drive motors **114** that engage the ring gear **112**. In addition to the various components that have been mentioned and shown in the Figures, it should also be appreciated that the access apertures **100** and general configuration of the rotor support frame **72** facilitates access to other internal generator components, for example: a radial rotor locking system which is provided for locking the rotor against radial movement e.g. during maintenance, an axial rotor locking system which is provided for locking the rotor against axial movement, a stray current protection system, rotary encoders which may be provided on the gearbox output shaft, other sensor systems such as accelerometers and temperature sensors that may be located on the gearbox output shaft and in other locations, brake disc components that are provided to apply a braking force to the rotor during maintenance, and pitch tube sealing parts which are provided to provide suitable sealing to pitch tube components that run centrally through the power train assembly to provide hydraulic and/or electrical power to a hydraulic pitching system located in the rotor hub.

(59) Considered as a whole, the configuration of the rotor support frame **72** provides a comparatively open volume within it which is sized and shaped beneficially to allow access to a person who can carry a range of maintenance activities. That volume can therefore be considered to define a service room inside the generator which is sized and shaped to accommodate service personnel. This is in contrast to known approaches in which access to various components is only achieved from outside the generator, or access from inside the generator is restricted due to the relatively positioning of internal components.

(60) Beneficially, the open volume inside of the generator rotor assembly **70** can be enclosed to provide a more protective environment yet still allow access to maintenance personnel when required. As shown in FIGS. **11a** and **11b**, the rear face of the generator service cabinet **81** may be provided with a closure **120** such as a door or hatch. The closure **120** may be a single door panel that covers an opening **121** in the generator cabinet **81** into the internal chamber in which the generator rotor assembly **70** is housed. Alternatively, the closure **120** may comprise a plurality of panels. As shown in the Figures, the closure **120** comprises two door panels **122,124**. Those door panels **122,124** may be mounted in any suitable way to allow opening. For example, the door panels **122,124** may be mounted by a fastening system that allows one or both of the panels to be lifted away from the cabinet opening when access inside the generator is required. Alternatively, one or both of the door panels **122,124** may be hinged such that they remain connected to the generator service cabinet **81** but can be moved to obtain access if required. FIG. **11b** shows one of the door panels **124** having a hinged fixing, and it will be appreciated that the other door panel **122** could also have such a fixing.

(61) Beneficially, the closure **120** encloses the internal volume of the generator which means that the internal environment is separated from the general interior of the nacelle within which it is installed.

(62) The closure **120** achieves a further benefit in that it can provide a single service entry point into the interior volume of the generator for servicing purposes. This means that the closure can be implemented with a single safety mechanism to permit controlled entry for servicing. The safety mechanism can be by way of keypad, finger scanner, or other access control technique. Preferably, access through the enclosure will only be allowed once the safety mechanism detects that one or more other safety measures or interlocks have been addressed—for example that the rotor has been locked so that the rotor is stationary, and that appropriate electrical systems have been rendered inactive so that there are no electrical risks associated with service personal accessing the interior of the generator. Since the interior volume inside the generator is provided as a servicing room with sufficient space, the area can be designed as a dedicated workspace with which to access a number of structures and systems associated with the generator, as discussed above. For example, a floor covering may be installed in the service room so that dropped items can be captured thereby avoiding risk of loose components inside the generator. Moreover, the closed environment of the service room, by virtue of the closure **120**, means that the accessible areas from the service room can be maintained in as clean a state as possible, which is easier to achieve on a small scale rather than requiring the confines of the nacelle to be sufficiently clean.

(63) Many modifications may be made to the specific examples described above without departing from the scope of the invention as defined in the accompanying claims. Features of one embodiment may also be used in other embodiments, either as an addition to such embodiment or as a replacement thereof.

Claims

1. A wind turbine powertrain assembly including a gearbox coupled to an electrical power generator for a wind turbine, wherein the power generator comprises a stator at a radially outward position and a rotor in a radially inward position, wherein the rotor comprises a cylindrical field structure that is configured to rotate about a generator rotor axis (R), and wherein the rotor is configured to define a central hollow portion, wherein the rotor further comprises a rotor support frame connected to and structurally supporting the cylindrical field structure, wherein the rotor support frame comprises: a rotor connection flange attaching the rotor support frame to the cylindrical field structure, a gearbox connecting flange connecting the rotor support frame to an output drive shaft of the gearbox, and a transition section extending between the rotor connection flange and the gearbox connection flange, the transition section defined at least in part by one or

- more stepping regions configured to provide access within the rotor support frame for a person standing therein, the or each stepping region shaped to define a surface that is inclined at an angle of less than 30 degrees to the rotor axis.
2. The wind turbine powertrain assembly of claim 1, wherein the or each stepping region provides a circumferentially-extending stepping surface extending about the rotor axis.
 3. The wind turbine powertrain assembly of claim 1, comprising a plurality of separate stepping regions.
 4. The wind turbine powertrain assembly of claim 3, wherein the plurality of separate stepping regions is separated by respective strengthening ribs.
 5. The wind turbine powertrain assembly of claim 1, wherein the transition section further includes an axially-facing end surface.
 6. The wind turbine powertrain assembly of claim 5, wherein the axially-facing end surface is generally aligned along the direction of the rotor axis (R) with the gearbox connection flange.
 7. The wind turbine powertrain assembly of claim 5, wherein the one or more stepping regions are positioned between the axially-facing end surface and the rotor connection flange, along the direction of the rotor axis (R).
 8. The wind turbine powertrain assembly of claim 5, wherein the axially-facing end surface comprises a plurality of access apertures.
 9. The wind turbine powertrain assembly of claim 8, wherein the plurality of access apertures each have an open area of at least 100 cm.^{sup.2}.
 10. The wind turbine powertrain assembly of claim 8, wherein the plurality of access apertures each have an open area of at least 200 cm.^{sup.2}.
 11. The wind turbine powertrain assembly of claim 1, wherein a distance (A) along the rotational axis between the gearbox connection flange and the rotor connection flange is between 20% and 60% of the internal diameter of the rotor connection flange.
 12. The wind turbine powertrain assembly of claim 1, wherein the rotor connection flange and the transition section form a first unitary component.
 13. The wind turbine powertrain assembly of claim 12, wherein the first unitary component is mechanically connected to the gearbox connection flange at a bolt ring radially inward of the axially-facing end surface.
 14. The wind turbine powertrain assembly of claim 1, further comprising a generator service cabinet that houses the rotor.
 15. The wind turbine powertrain assembly of claim 14, wherein the generator service cabinet is connected to a gearbox housing.
 16. The wind turbine powertrain assembly of claim 14, wherein the generator service cabinet defines an internal chamber which houses the rotor, and wherein the generator service cabinet is provided with one or more closure elements which are configured to selectively close an opening defined by the generator service cabinet into the internal chamber.
 17. The wind turbine powertrain assembly of claim 1, wherein the gearbox includes at least one planetary gear stage.
 18. The wind turbine powertrain assembly of claim 1, wherein a distance (A) along the rotational axis between the gearbox connection flange and the rotor connection flange is between 20% and 40% of the internal diameter of the rotor connection flange.
 19. The wind turbine powertrain assembly of claim 1, wherein the rotor connection flange and the transition section form part of a first unitary component and the gearbox connection flange forms part of a second unitary component coupled to the first unitary component.
 20. A wind turbine comprising a wind turbine powertrain assembly according to claim 1.
-